🌊 HydroGraphRAG: Intelligent GIS Assistant

Geospatial code generation and visualization based on knowledge graphs.

Enter hydrology query:

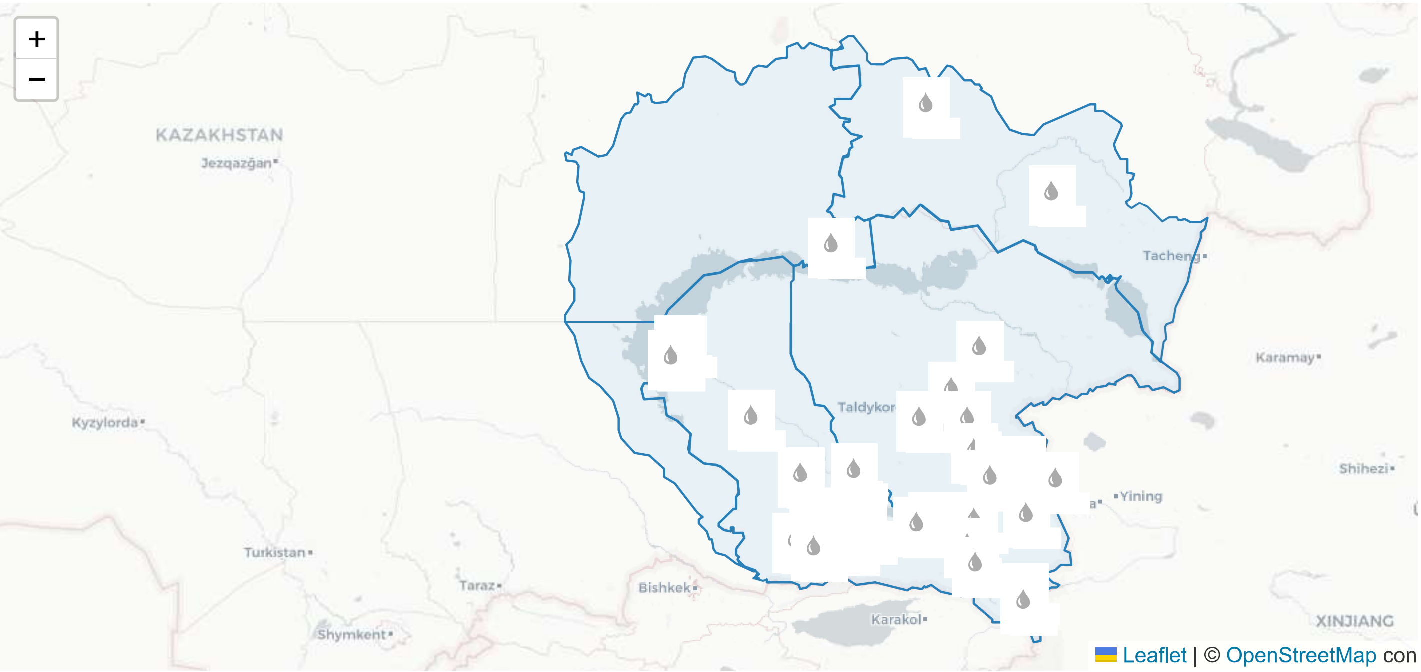
Describe me detailed the water statistics of Almaty Region and Almaty City. What rivers does it have?

Generate solution

✅ Solution generated!

Category	Triples extracted	Points on map
SEMI-EXPLICIT	511	77

🗺️ Geospatial Visualization



📝 Analytical Report and Code

Modeling Solution:

The Almaty Region and Almaty City are located in the central part of Kazakhstan, with a significant presence of rivers. The Kishi Almaty River is one of the most prominent rivers in the region, flowing through Almaty city. Additionally, there are several other smaller rivers that contribute to the water resources of the area.

The water quality in Almaty City is classified as class 2, indicating a good level of water purity and suitability for human consumption. The basin area of the Kishi Almaty River is approximately 520 km², with a length of about 32 km. The river's width at its mouth is around 43.15 meters.

Implementation Code:

```
import geopandas as gpd
import folium
from shapely import wkt

# 1. Load basin
basin_data = gpd.read_file(r"data/basin_data.shp").to_crs('EPSG:4326')
centroid = basin_data.geometry.centroid.iloc[0]

# 2. Init map
m = folium.Map(location=[centroid.y, centroid.x], tiles='CartoDB positron', zoom_start=8)
folium.GeoJson(basin_data.to_json(), style_function=lambda x: {'fillColor': 'green', 'color': 'darkgreen', 'fillOpacity': 0

# 3. Add Graph Knowledge points
points = [
    {"name": "Kishi Almaty River - M Mynzhilky", "wkt": "POINT(76.786 43.099021)"},
    {"name": "Kishi Almaty River - Tuyuksu Alpine base", "wkt": "POINT(77.078147 43.084405)"},
    {"name": "Kishi Almaty River - (near Almaty)", "wkt": "POINT(77.0331 43.0927)"},
    {"name": "Kishi Almaty River - Almaty city", "wkt": "POINT(76.54 43.15)"},
]

# 4. Draw markers
for p in points:
    try:
        geom = wkt.loads(p["wkt"])
        folium.Marker(location=[geom.y, geom.x], popup=p["name"]).add_to(m)
    except Exception:
        pass

m.save("api_1780742557.html")
```